Arquitectura de Robots

The Geometry of Robots pt 2

Dr. Mario Angel Rico Mendez

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FIME
FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA



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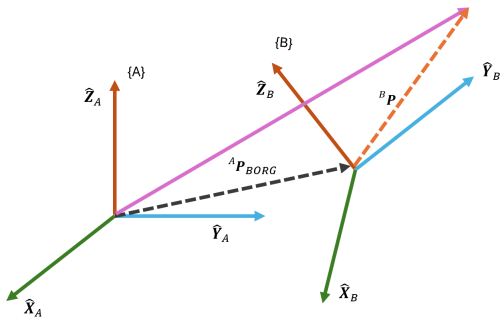
Transformations with Arbitrary Frames

We will now examine the general scenario of mapping.

The origin of frame B does not coincide with the origin of frame A and is offset by a general vector.

Also B is rotated with respect to A

Figura 1: *General transform of a vector.*



Transformations rotation and translation

We can first change B^P to its description relative to an intermediate frame that has the same orientation as A.

Next, consider the shift between origins by performing basic vector addition.

$$ {}^{A}P = {}^{A}_{B}R\, {}^{B}P + {}^{A}P_{BORG} \tag{1} $$

This equation represents a general transformation that maps a vector's description from one frame to another frame.

Homogeneous transform.

To express the mathematics from the previous equation in the proposed matrix operator form, we introduce a 4×4 matrix operator and employ 4×1 position vectors, giving it the structure

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}R & {}^{A}P_{BORG} \\ 000 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} \tag{2}$$

In other words,

1. a ‘‘1’’ is added as the last element of the 4×1 vectors;
2. a row ‘‘[0 0 0 1]’’ is added as the last row of the 4×4 matrix

The 4×4 matrix is called a **homogeneous transform.**

Homogeneous coordinates and matrices

1 Transformations with Arbitrary Frames

The representation of the location of solids in a dimensional space using homogeneous coordinates is carried out using coordinates of an (n+1)-dimensional space.

An n-dimensional space is represented in homogeneous coordinates by (n+1) dimensions, such that a vector $p(x, y, z)$ is represented by $p(wx, wy, wz, w)$ where w has an arbitrary value and represents the scale.

$$p = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} xw \\ yw \\ zw \\ w \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \tag{3}$$

Example

The vector 2i + 3j + 4k can be expressed in homogeneous coordinates as [2,3,4,1] or also as [-6,-9,-12,-3].

$$p = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} xw \\ yw \\ zw \\ w \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(1) \\ 3(1) \\ 4(1) \\ 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

A homogeneous transformation matrix T is defined as a 4×4 matrix that represents the transformation of a homogeneous coordinate vector from one coordinate system to another.

General transform

For our objectives, it can be seen simply as a framework to express the rotation and translation of the general transformation in a unified matrix format.

$$ {}^{A}P = {}^{A}_{B}T\, {}^{B}P \tag{4} $$

The perspective is the term ”000” and the scaling is ”1”, this is when the last row is ‘‘[0 0 0 1]’’ .

The description of frame B relative to A is ${}^{A}_{B}T$. The description encompasses both the rotation and translation of the frame.

Section 1.1

Example

Example

1 Transformations with Arbitrary Frames

Figure 3 shows a frame B, which is rotated relative to frame A about Z by 30 degrees, translated 10 units in X_A, and translated 5 units in Y_A. Find ${}^{A}P$, where ${}^{B}P = [3.0, 7.0, 0.0]^T$.

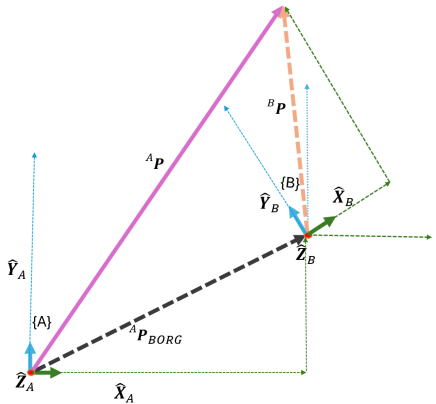


Figura 2: *Frame B rotated and translated.*

Example solution

The definition of frame B is

$$ {}^{A}_{B}T = \begin{bmatrix} {}^{A}_{B}R & {}^{A}P_{BORG} \\ 000 & 1 \end{bmatrix} = \begin{bmatrix} \cos 30 & -\sin 30 & 0 & 10 \\ \sin 30 & \cos 30 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Given

$$ {}^{B}P = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix} $$

Example solution cont ...

we use the definition

$$ {}^{A}P = {}^{A}_{B}T\,{}^{B}P $$

$$ {}^{A}P = \begin{bmatrix} \cos 30 & -\sin 30 & 0 & 10 \\ \sin 30 & \cos 30 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} (\cos 30 \cdot 3) + (-\sin 30 \cdot 7) + (0 \cdot 0) + (10 \cdot 1) \\ (\sin 30 \cdot 3) + (\cos 30 \cdot 7) + (0 \cdot 0) + (5 \cdot 1) \\ (0 \cdot 3) + (0 \cdot 7) + (1 \cdot 0) + (0 \cdot 1) \\ (0 \cdot 3) + (0 \cdot 7) + (0 \cdot 0) + (1 \cdot 1) \end{bmatrix} = $$

$$ {}^{A}P = \begin{bmatrix} 9.098 \\ 12.562 \\ 0 \\ 1 \end{bmatrix} $$

Example solution ${}^{A}P$

1 Transformations with Arbitrary Frames

$$ {}^{A}P = \begin{bmatrix} 9.098 \\ 12.562 \\ 0 \\ 1 \end{bmatrix} $$

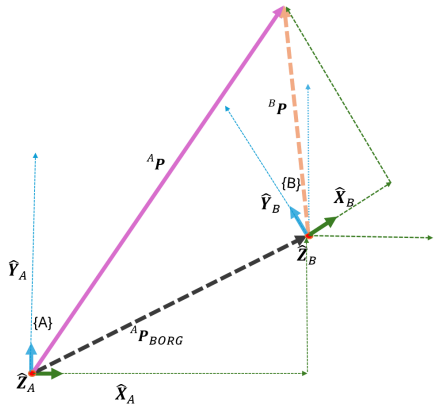


Figura 3: *Frame B rotated and translated.*

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- Transformations with Arbitrary Frames
- Operators: translations, rotations and tranformation

Section 2.1

Translational operators

Operators

2 Operators: translations, rotations and tranformation

The mathematical expressions utilized to map points across different frames can also be seen as **operators** that perform **translations, rotations**, or a combination of both on points and vectors.

Translational operators

A translation shifts a point in space by a specified distance in a particular vector direction.

The figure illustrates the translation of vector ${}^{A}P_1$ by vector ${}^{A}Q$ in a pictorial manner. The vector ${}^{A}Q$ provides the requisite information to execute the translation.

The result of the operation is a new vector ${}^{A}P_2$, calculated as

$$ {}^{A}P_2 = {}^{A}P_1 + {}^{A}Q $$

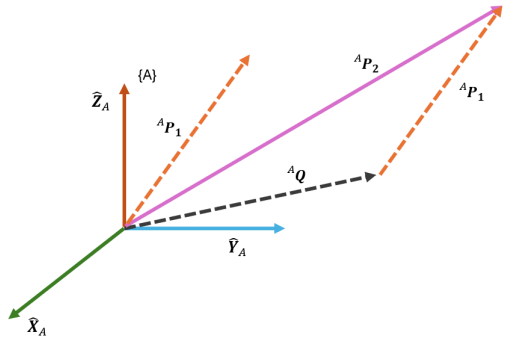


Figura 4: *Translation operator.*

Translation operator x,y,z

To write this translation operation as a matrix operator, we use the notation

$$^{A}P_2 = D_Q(q)\,^{A}P_1 \tag{5}$$

where q is the signed magnitude of the translation along the vector direction Q. The D_Q operator may be thought of as a homogeneous transform of a special simple form:

$$D_Q(q) = \begin{bmatrix} 1 & 0 & 0 & q_x \\ 0 & 1 & 0 & q_y \\ 0 & 0 & 1 & q_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

where q_x, q_y, and q_z are the components of the translation vector and $q = \sqrt{q_x^2 + q_y^2 + q_z^2}$

Section 2.2

Rotational operators

Rotational operators

Another interpretation of a rotation matrix is as a **rotational operator** that operates on a vector AP1 and changes that vector to a new vector, AP2, by means of a rotation, R.

That is, we may write

$$^{A}P_2 = R\,^{A}P_1 \tag{7}$$

Rotation matrix is easily viewed as an operator,

$$^{A}P_2 = R_Z(\theta)\,^{A}P_1 \tag{8}$$

Rotates about the Z

This operator can be written as a homogeneous transform whose position-vector part is zero.

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{9}$$

Section 2.3

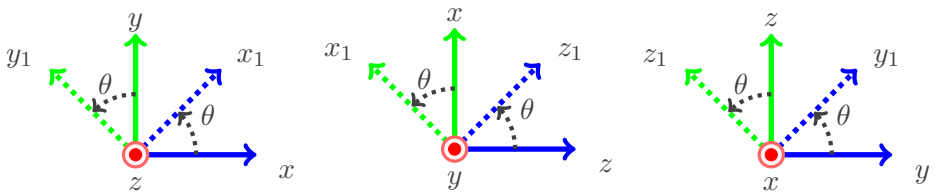
Rotations

Rotations

2 Operators: translations, rotations and tranformation

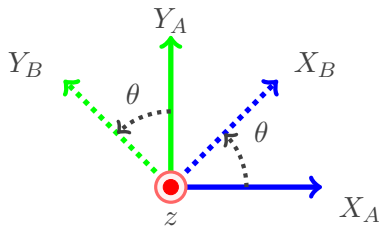
A rotation of the coordinate axes can be decomposed into rotations of each of the axes

The transformations of each of these rotations are given by $\mathbf{R}_Z$, $\mathbf{R}_Y$ and $\mathbf{R}_X$



Z-Rotation Matrix

$$\mathbf{R}_Z = \overbrace{\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}}^{\begin{matrix} X_B & Y_B & Z_B \end{matrix}} \left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} \begin{matrix} X_A \\ Y_A \\ Z_A \end{matrix} \quad (10)$$

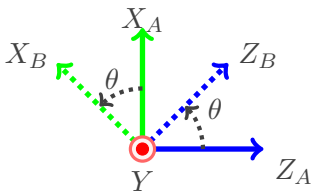


dot product $|A||B|\cos\theta$
$\sin\theta = \cos(90 - \theta)$

Y-Rotation Matrix

$$\mathbf{R}_Y = \overbrace{\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}}^{\begin{matrix} X_B & Y_B & Z_B \end{matrix}} \left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} \begin{matrix} X_A \\ Y_A \\ Z_A \end{matrix}$$

(11)

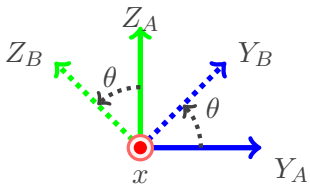


dot product $|A||B|\cos\theta$
$\sin\theta = \cos(90 - \theta)$

X-Rotation Matrix

$$\mathbf{R}_X = \overbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}}^{\begin{matrix} X_B & Y_B & Z_B \end{matrix}} \left.\begin{matrix} \\ \\ \\ \end{matrix}\right\} \begin{matrix} X_A \\ Y_A \\ Z_A \end{matrix}$$

(12)



dot product $|A||B|\cos\theta$
$\sin\theta = \cos(90 - \theta)$

Section 2.4

Transformation operators

Transformation operators

2 Operators: translations, rotations and tranformation

A frame has another interpretation as a transformation operator.

The operator T rotates and translates a vector A_1^P to compute a new vector,

$$ {}^{A}P_2 = {}^{A}_{B}T\,{}^{A}P_1 \tag{13} $$

A transform is usually thought of as being in the form of a homogeneous transform with general rotation-matrix and position-vector parts.

Section 2.5

Example

Example

2 Operators: translations, rotations and tranformation

Figure shows a vector ${}^{A}P_1$. We wish to rotate it about Z by 30 degrees and translate it 10 units in X_A and 5 units in Y_A. Find ${}^{A}P_2$, where ${}^{A}P_1 = [3.0 \quad 7.0 \quad 0.0]^T$.

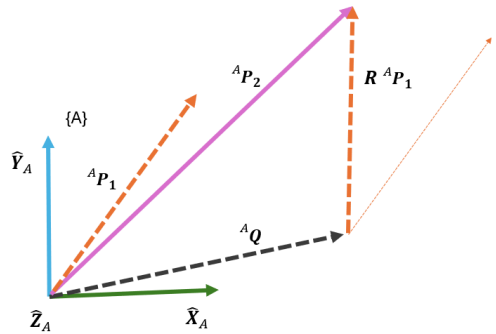


Figura 5: *The vector ${}^{A}P_1$ rotated and translated to form ${}^{A}P_2$.*

Example Solution

2 Operators: translations, rotations and tranformation

The operator T , which performs the translation and rotation, is

$$ {}^{A}_{B}T = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & Q_x \\ \sin\theta & \cos\theta & 0 & Q_y \\ 0 & 0 & 1 & Q_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.50 & 0 & 10 \\ 0.50 & 0.866 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ {}^{A}P_2 = {}^{A}_{B}T \ {}^{A}P_1 = \begin{bmatrix} 0.866 & -0.50 & 0 & 10 \\ 0.50 & 0.866 & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.09 \\ 12.56 \\ 0 \\ 1 \end{bmatrix} $$

Example solution

2 Operators: translations, rotations and tranformation

$$ {}^{A}P_2 = {}^{A}_{B}T \ {}^{A}P_1 = \begin{bmatrix} 9.09 \\ 12.56 \\ 0 \\ 1 \end{bmatrix} $$

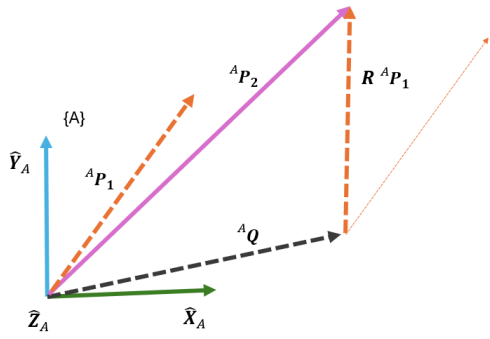


Figura 6: *The vector ${}^{A}P_1$ rotated and translated to form ${}^{A}P_2$.*

Summary

2 Operators: translations, rotations and tranformation

The text outlines the introduction of concepts starting with translation only, followed by rotation only, and eventually covering the general case of rotation about a point combined with the translation of that point. Understanding the general case encompasses the simpler cases of translation and rotation.

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Q&A

Thank you for your attention

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